

Subject Description Form

Subject Code	LSGI4220
Subject Title	Urban Remote Sensing
Credit Value	3
Level	4
Pre-requisite	LSGI2223 Geographic Information Science
Objectives	• To give students a basis to apply remote sensing methods and techniques to practical problems of urban and built-up lands, environments, and ecosystems. • To enable students to jointly use remote sensing and GIS methods and techniques to help understand urbanization processes and to address issues of the environmental and ecological impacts induced by the urbanization.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1 Have a working knowledge of urbanization concepts and processes (L1) 2 Have a working knowledge of current urban environmental issues and problems (L2) 3 Have a working knowledge of urban climate and ecosystem issues (L3) 4 Understand the current applications of the remote sensing methods and techniques to urban land resources in Hong Kong and the Greater Bay Area (L4) 5 Solve problems of the adverse environmental and ecological impacts caused by urbanization using remote sensing skills (L5) 6 Integrate remote sensing and GIS methods to solve urbanization and urban environmental and ecological issues (L6)
Subject Synopsis/ Indicative Syllabus	A. Remote sensing and GIS integration B. Urban land use, and impervious surfaces C. Urban landscapes and planning considerations D. Urban heat island and air quality E. Urban flooding, landslide, and other hazards F. Urban population estimation and quality of life assessment G. Public health in urban environments H. Cities at night

Teaching/Learning Methodology	Teaching and learning materials will be delivered on-line for students to download easily. Contact hours will be used for formal lectures, in-class discussions and presentations, and lab exercises. A group project will then follow to develop students’ high-level cognitive understanding and integration of knowledge.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			1	2	3	4	5	6
	1. Written examination	40	√	√	√	√	√	√
	2. Lab exercises	40				√	√	√
	3. Group project	20	√	√	√	√	√	√
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: • Students will be assessed of their remote sensing and GIS integrated techniques in lab sessions, by submission of practical reports and individual project report. • The group project is to assess students’ understanding of and to use the remote sensing-GIS integration methods and techniques acquired in this subject to solve urban land, environmental, ecological, and socioeconomic issues. • The end of semester examination is to test students’ understanding of the theories, techniques, and methods of the remote sensing-GIS integration.							

Student Study Effort Expected	Class contact:	
	▪ Lectures	26 Hrs.
	▪ Lab exercises	26 Hrs.
	Other student study effort:	
	▪ Reading of journal papers	23 Hrs.
	▪ Assignment completion and report writing	30 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	Texts: • Weng, Q. 2009. Remote Sensing and GIS Integration: Theories, Methods, and Applications. New York: McGraw Hill Professional. • Weng, Q. 2019. Techniques and Methods in Urban Remote Sensing, IEEE Press / Wiley. • Weng, Q. 2012. An Introduction to Contemporary Remote Sensing. McGraw-Hill Professional. Journals: 1. Computers, Environment and Urban Systems 2. Landscape and Urban Planning 3. GIScience and Remote Sensing 4. Journal of Environmental Management 5. ISPRS Journal of Photogrammetry and Remote Sensing 6. Remote sensing of Environment	